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CS 470 Final Reflection

In tis course, I was able to learn valuable skills in regards to understanding features of a cloud environment, along with managing and deploying a full stack application into the cloud. I learned about how I can create a serverless environment which allows me to not have to manage the physical or virtual hardware and can focus more on other production tasks. A serverless environment cant hen scale for me and provide necessary resources that I need to help me manage costs. The AWS Lambda service allows me to use the resources based on if any events are actually occurring. Also, I learned about hot resources like the files of my application are stored in a objects referred to as buckets. This design allows me to skip physical hard drives and also do not have depend on backups and relying on RAID to copy my data redundantly. This also has the feature that scales to allow me to control my costs. In addition, I learned the strengths of using a NoSQL database like Dynamo which is non relational. It is schema less and organized through the use of key-value document structures. I don't have to spend a lot of time provision the database either. I can relate my information through tables, even if the information doesn't have anything to do with each other. Furthermore, I learned more about securing the cloud environment which is vital. There are ways to setup roles and policies which restrict access to the environment when it comes to users or triggered events against. I can set rules against everything in the infrastructure managed either in groups or individual users. Lastly, understanding how to build an application in a container using Docker compose allows the ability to control and manage multiple features of an app one file, and start them in one command. This manages pieces of an application like the backend and frontend of an application. A containerized app also helps being able to deploy the application to different environments without having to restructure the application to run on different environments.The command line steps are easy to use in order to deploy an app locally to the cloud environment and then configuring it to be public.

Understanding how to build full stack applications is in demand, but the leverage in knowing how to deploy and manage in a cloud environment like AWS takes development to another level. Knowing this skills will help me lead into a more complicated roles like principal and staff engineer. Knowing the cloud environment also helps understand how to operate a business for cost and efficiency. When it comes to system design, knowing how to handle cost, will help to build a environment that is not wasting resources. Understanding the type of application that is being built will allow me to control my cost because I would know which direction I need to take in regard to any additional configuration and cost. Elasticity provides the resources I need only when I need them, so I don't have to pay for parts of the infrastructure that is not being used. Elasticity within AWS knows how to handle adding and removing resources based on understanding the environment. Pay for service is a method that bills based on what is actually used. You are not subscribing to a plan for different tiers without worrying about overspending on computing that is not being used.